



CS-860: ARTIFICIAL **INTELLIGENCE**

- **Environments and Agents**
 - **Environment Categories**
 - **What is an AI Agent?**
 - **Types of Agents**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

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Environments and Agents



• Environment Categories

Environment Categories

- Fully Observable vs. Partially Observable
- Single-Agent vs. Multi-Agent
- Deterministic vs. non-deterministic
- Episodic vs. Sequential
- Static vs. Dynamic
- Discrete vs. Continuous
- Known vs. Unknown



Environments and Agents



- **Environment Categories**
 - **Fully Observable vs Partially Observable**

Fully Observable vs. Partially Observable

- If an agent's sensors give it access to the complete state of the environment at each point in time => Fully observable
- Sensors detect all aspects that are relevant to the **choice of action** which depends on the performance measure.
- Do not need to maintain any internal state to keep track of the world.
- If an agent's sensors do not give it access to the complete state of the environment because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
- Vacuum agent with only local dirt sensor cannot tell whether there is dirt in other squares.
- An automated taxi agent cannot see what other drivers are thinking.



Environments and Agents



- **Environment Categories**
 - Fully Observable vs Partially Observable
 - **Single Agent vs Multi Agent**

Single Agent vs Multi Agent

- Self explanatory
- Agent solving a crossword puzzle by itself is in single agent environment
- Agent playing chess is in a two-agent environment.
- Automated Taxi Agent Example
- Does Taxi, agent 'A', have to treat an object (vehicle) as agent 'B'?
- The key distinction is whether B's behavior is best described as maximizing a performance measure whose value depends on agent A's behavior.
- Chess Example => Competitive Multi-agent Environment
- Taxi Example => Partially Cooperative Multi-agent Environment
- Agent-design problems in multi-agent environments.



Environments and Agents



- **Environment Categories**
 - Fully Observable vs Partially Observable
 - Single Agent vs Multi Agent
 - **Deterministic vs Non-Deterministic**

Deterministic vs Non-Deterministic

- If the next state of the environment is completely determined by the current state and the action executed by an agent
- If the next state of the environment is not completely/partially determined.
- Most real-life situations are so complex that it is impossible to keep track of all the unobserved aspects, they are treated as nondeterministic.
- Taxi driving is nondeterministic.
- Vacuum Agent example environment is deterministic.
- Difference between stochastic vs. nondeterministic
- A model of the environment is stochastic if it explicitly deals with probabilities (e.g., “there is a 25% chance of rain tomorrow”) and nondeterministic if the possibilities are listed without being quantified (e.g., “there is a chance of rain tomorrow”)



Environments and Agents



- **Environment Categories**

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- Deterministic vs Non-Deterministic
- **Episodic vs Sequential**

Episodic vs Sequential

- The agent's experience is divided into episodes. In each episodes, the agent receives a percept and then performs a single action. The next episode does not depend on the action taken on the previous episodes.
- In sequential environments, the current decision could affect all future decisions.
- An agent that has to spot defective parts on an assembly line bases each decision on the current part, regardless of previous decisions.
- Chess and Taxi examples, both are sequential: short term actions can have long term consequences.
- Episodic environments are much simpler. Agent does not need to think ahead



Environments and Agents



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- **Static vs Dynamic**

Static vs Dynamic

- If an environment can change while an agent is deliberating, then the environment is termed as dynamic.
- Otherwise, it is a static environment.
- Static environments are easy to deal
- Dynamic environments are continuously asking the agent what it wants to do if the agent has not decided.
- If the environment is static but the performance score of the agent is changing over time, the environment is termed as semi-dynamic.
- Chess Example => Semi-dynamic when played with a clock
- Taxi Example => Dynamic
- Crossword Puzzle => Static



Environments and Agents



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 - Static vs Dynamic
 - **Discrete vs Continuous**

Discrete vs Continuous

- The continuous/discrete distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agents.
- Chess => Discrete (finite states, discrete percepts and actions)
- Taxi driving => Continuous (Why??)
- Continuous sequence, speed, time, location of the taxi
- Actions: steering, braking, acceleration
- Percepts: cameras (can be treated as discrete but normally continuous feed with varying intensity and locations)



Environments and Agents



- **Environment Categories**
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 - Discrete vs Continuous
 - **Known vs Unknown**

Known vs Unknown

- This distinction refers not the environment itself but to the agent's state of knowledge about the “laws of physics” of the environment.
- It is a known environment if the outcomes (or outcome probabilities of the environment is nondeterministic) for all actions are given.
- If the environment is unknown, then the agent must learn how it works in order to make good decisions.
- Not similar to fully observable vs. partially observable
- Known environment can be partially observable => solitaire game
- Unknown environment can be fully observable => in a new video game, the screen show the entire game state, but you still don't know what the buttons until you try them.



Environments and Agents



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 - Known vs Unknown
 - **Examples**

Examples of Task Environments

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword Puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	nondeterministic	Sequential	Static	Discrete
Backgammon	Fully	Multi	nondeterministic	Sequential	Static	Discrete
Taxi Driving	Partially	Multi	nondeterministic	Sequential	Dynamic	Continuous
Medical Diagnosis	Partially	Single	nondeterministic	Sequential	Dynamic	Continuous
Image Analysis	Fully	Single	deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	nondeterministic	Episodic	Dynamic	Continuous
Refinery Controller	Partially	Single	nondeterministic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	nondeterministic	Sequential	Dynamic	Discrete



Environments and Agents



- Environment Categories
- What is an AI Agent?
 - Structure of Agents

Structure of Agents

- Agent Program
 - Agent function
 - Agent Architecture
-
- The job of AI is to design an agent program that implements the agent function, i.e., the mapping from percepts to actions.
 - The agent program runs on a computing device with sensors and actuators. This is called agent architecture.

agent = architecture + program



Environments and Agents



- Environment Categories
- **What is an AI Agent?**
 - Structure of Agents
 - **Table Driven Agent**

Table Driven Agent

- **Agent Program**

Figure 2.7

function TABLE-DRIVEN-AGENT(*percept*) **returns** an action

persistent: *percepts*, a sequence, initially empty

table, a table of actions, indexed by percept sequences, initially fully specified

append *percept* to the end of *percepts*

action $\leftarrow$ LOOKUP(*percepts*, *table*)

return *action*

The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.



Environments and Agents

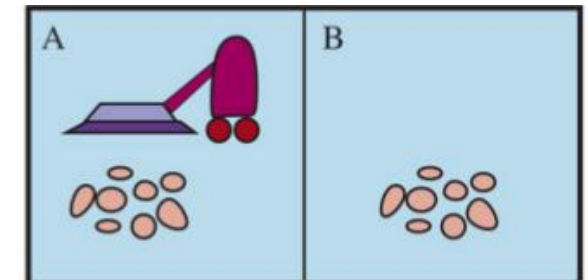


- Environment Categories
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Table Driven Agent - Example

- **Robotic Vacuum-Cleaning Agent**
- A vacuum cleaner world with just two locations. Each can be clean or dirt. Agent can move either right or left and can clean the square.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮



Environments and Agents



- Environment Categories
- What is an AI Agent?
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Table Driven Agent

- Agent Program

Figure 2.7

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
               table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

- Let P be the possible percepts and let T be the lifetime of the agent (the total number of percepts it will receive).
- The lookup table will contain entries

$$\sum_{t=1}^T |P|^t$$



Environments and Agents



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Table Driven Agent

- Consider the automated taxi: the visual input from a single camera comes in at the rate of 30 frames per second, 1080x720 pixels with 24bits of color information.
- This gives a lookup table with over $10^{600,000,000,000}$ entries for an hour's driving.
- In case of Chess, 10^{150} entries
- The daunting size of these tables means
 - a) no physical agent in this universe will have the space to store the table;
 - b) the designer would not have time to create the table; and
 - c) no agent could ever learn all the right table entries from its experience.



Environments and Agents



- Environment Categories
- What is an AI Agent?
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Agent Programs

- The key challenge of AI is to find out how to write program that, to the extent possible, produce rational behavior from a smallish program rather than from a vast table.
- Four basic kinds of agent programs that embody the principles underlying almost all intelligent systems:
 - Simple Reflex Agents
 - Model-Based Reflex Agents
 - Goal Based Agents
 - Utility Based Agents



Environments and Agents



- Environment Categories
- What is an AI Agent?
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 - Simple Reflex Agents

Simple Reflex Agents

- Simplest kind of Agents
- Agents select actions on the basis of the current percept
- Ignores the rest of the percept history
- Vacuum Agent example from the last lecture

Figure 2.8

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action  
  
    if status = Dirty then return Suck  
    else if location = A then return Right  
    else if location = B then return Left
```

The agent program for a simple reflex agent in the two-location vacuum environment. This program implements the agent function tabulated in [Figure 2.3](#).



Environments and Agents



- Environment Categories
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Simple Reflex Agents

- Simple reflex behaviors occur in even complex environments
- Taxi Driver example
- If the car in front brakes and its brake lights come on, then you should notice this and initiate braking.
- A Condition-Action Rule
 - If car-in-front-is-braking then initiate-braking
- Another example is Human eye blinking.



Environments and Agents

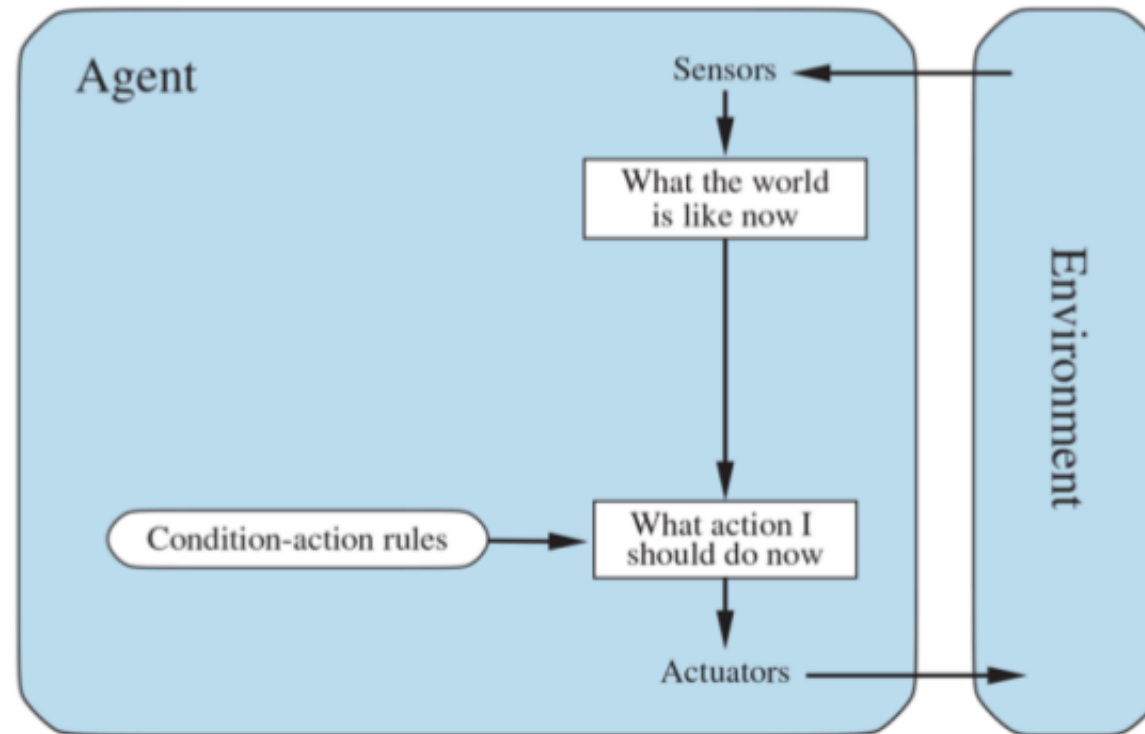


- Environment Categories
- What is an AI Agent?
- **Types of Agents**
 - Simple Reflex Agents

Simple Reflex Agents

- Structure of general program of simple reflex agents in schematic form

Figure 2.9



Schematic diagram of a simple reflex agent. We use rectangles to denote the current internal state of the agent's decision process, and ovals to represent the background information used in the process.



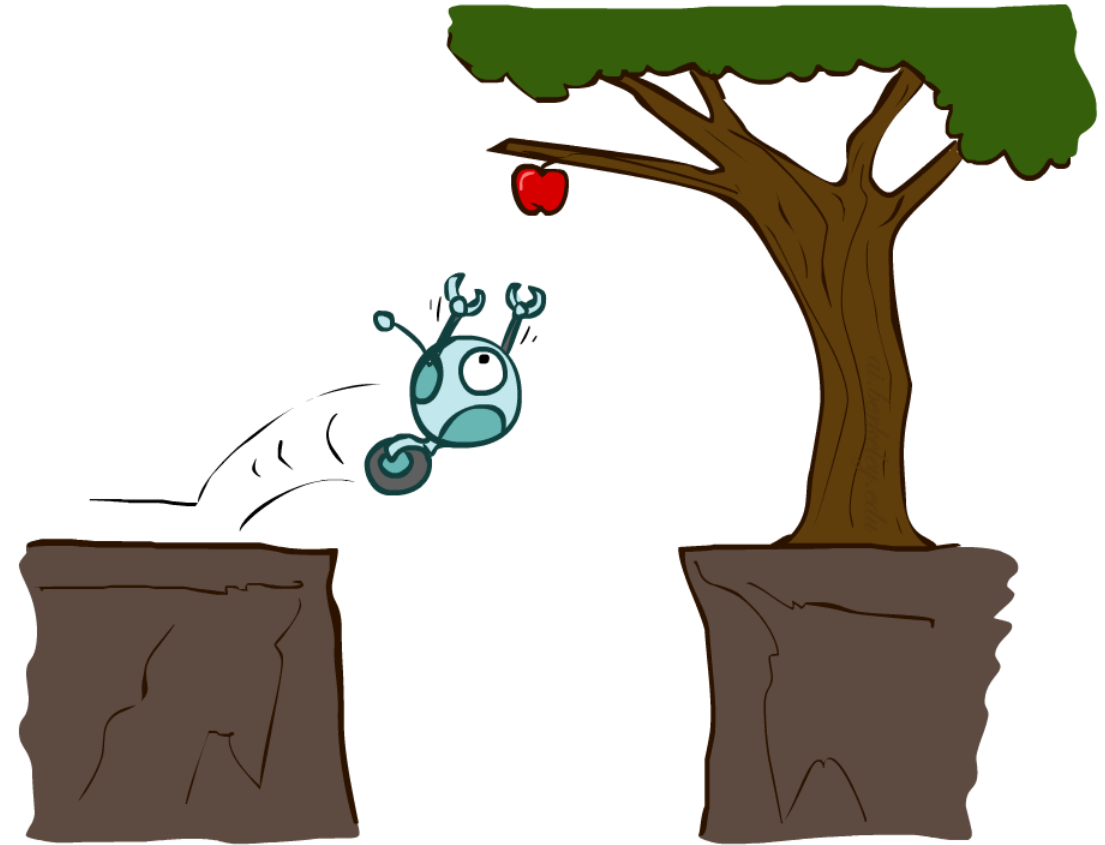
Environments and Agents



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Simple Reflex Agents

- Choose action based on current percept (and maybe memory)
- May have memory or a model of the world's current state
- Do not consider the future consequences of their actions
- **Consider how the world IS**



Can a reflex agent be rational?



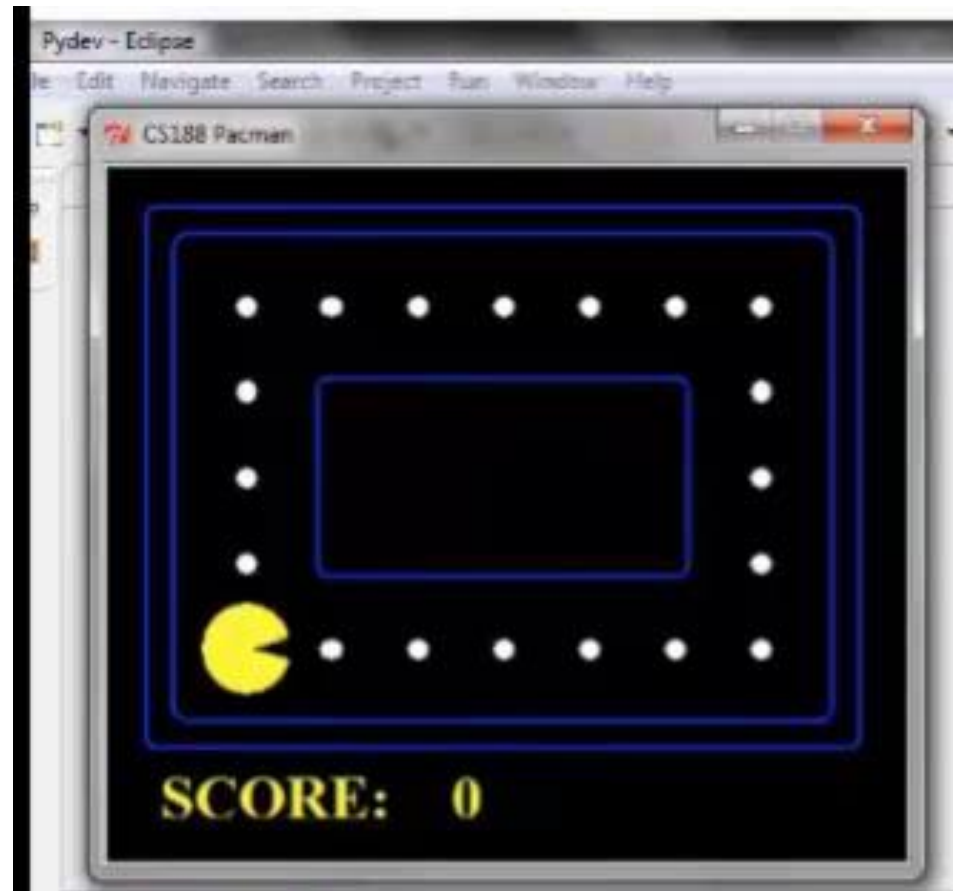
Environments and Agents



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Simple Reflex Agents

Video of Demo Reflex Optimal



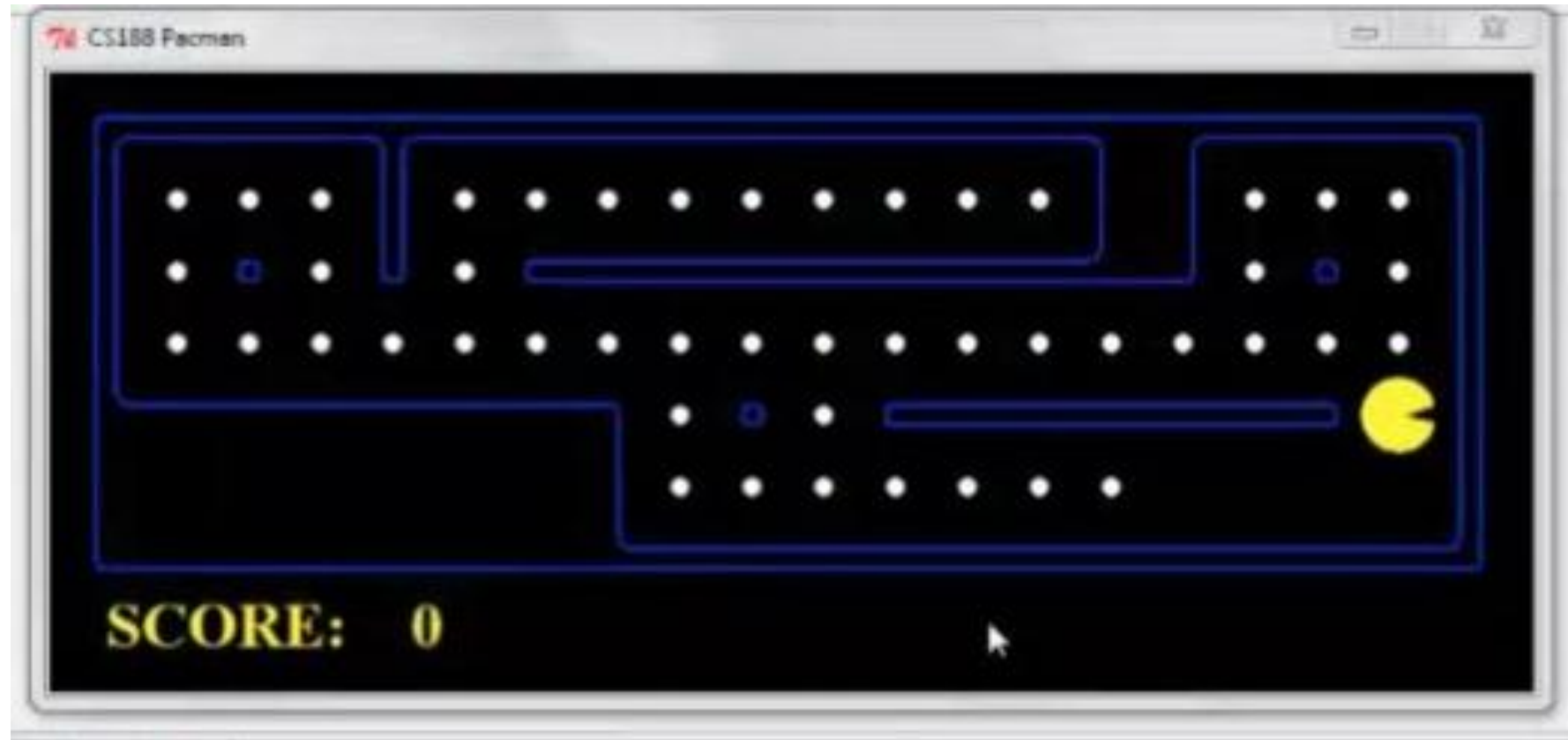
Environments and Agents



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Simple Reflex Agents

Video of Demo Reflex Odd



Environments and Agents



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Simple Reflex Agents

- General agent program of simple reflex agents

Figure 2.10

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

- Agent will work only if the correct decision can be made on the basis of just the current percept.
- Only if the environment is fully observable.
- Car braking Example. Vacuum Agent Example.



Environments and Agents



- Environment Categories
- What is an AI Agent?
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 - Simple Reflex Agents
 - **Model-Based Reflex Agents**

Model-Based Reflex Agents

- Can handle partial observability
- Keep track of the part of the world it can not see now.
- Maintaining internal state, which consists of percept history.
- Car braking problem. What would be the internal state?
- Previous frame from the camera
- Car Lane Changing Problem. What would be the internal state?
- Keeping track of other vehicles if the agent can not see them.



Environments and Agents



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Model-Based Reflex Agents

- Internal State
- Updating the internal state information requires two kinds of knowledge to be encoded in the agent program.
 1. How the world changes over time?
 - i. Effects of the agent's action
 - When the agent turns the car steering clockwise, the car turns to the right.
 - ii. How the world evolves independently of the agent.
 - When its raining, the car's cameras can get wet.
- This knowledge about “how the world works” when implemented using scientific theories is called a transition model of the world.



Environments and Agents



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- What is an AI Agent?
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 - **Goal-Based Agents**

Goal-Based Agents

- Knowledge about the current state is not enough.



- Correct decision depends on where the car is trying to get to.
- Apart from current state description of the world, the agent also needs some sort of a goal information that describes situations that are desirable.
- The agent program can combine this goal information with the model (transition + sensor) to choose actions that achieve the goal.

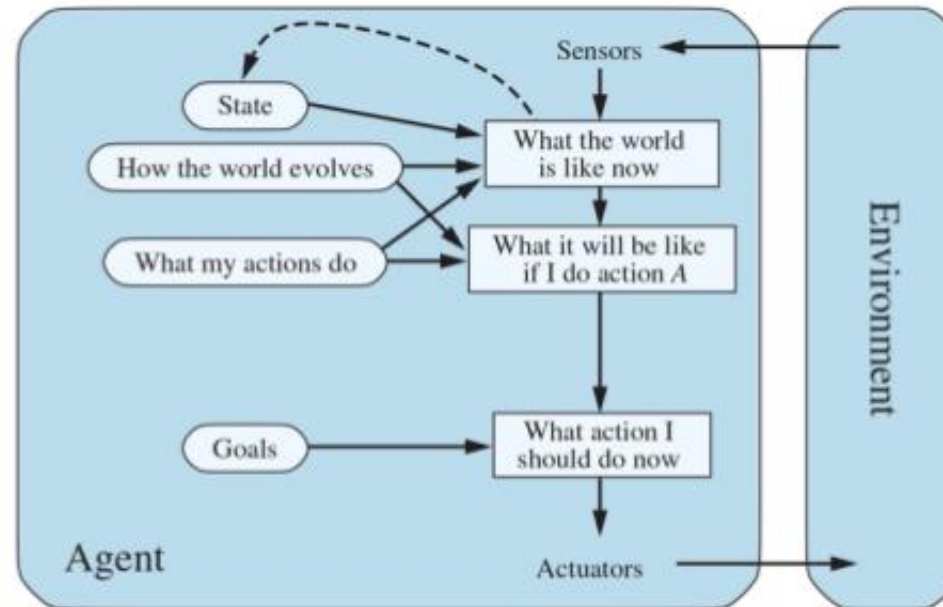


Environments and Agents



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Goal-Based Agents



- The decision making in goal-based agents is different from the condition-action rules since it involves the consideration of future “What will happen if I do such-and-such?” and “Will that make me happy?”
- This information is not represented in reflex agent designs because they use built-in rules mapped directly from percepts to action.



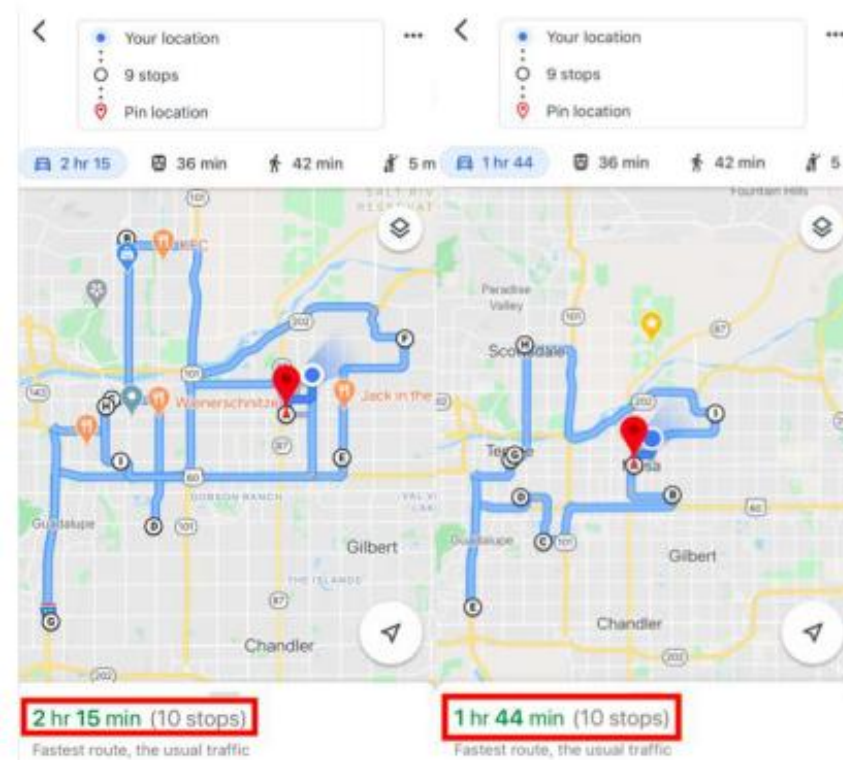
Environments and Agents



- Environment Categories
- What is an AI Agent?
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 - Goal-Based Agents
 - **Utility-Based Agents**

Utility-Based Agents

- Goals alone are not enough to generate high-quality behavior.



- Quicker, Safer, Cheaper, Reliable.
- Goals provide a binary distinction between happy and unhappy.



Environments and Agents



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Utility-Based Agents

- A general performance measure should be the comparison of how happy different world states would make the agent.
- From now on, Happy $\Rightarrow$ Utility (more scientific)
- An agent's utility function is essentially an internalization of the performance measure provided the internal utility function and the external performance measure are in agreement.
- Consider a case, when there are conflicting goals, only some of which can be achieved. E.g., speed and safety.
- Utility function specifies the appropriate trade-off.
- Consider another case, when there are several goals that the agent can aim for, none of which can be achieved with certainty.
- Utility function provides a way in which the likelihood of success can be weighed against the importance of the goals.



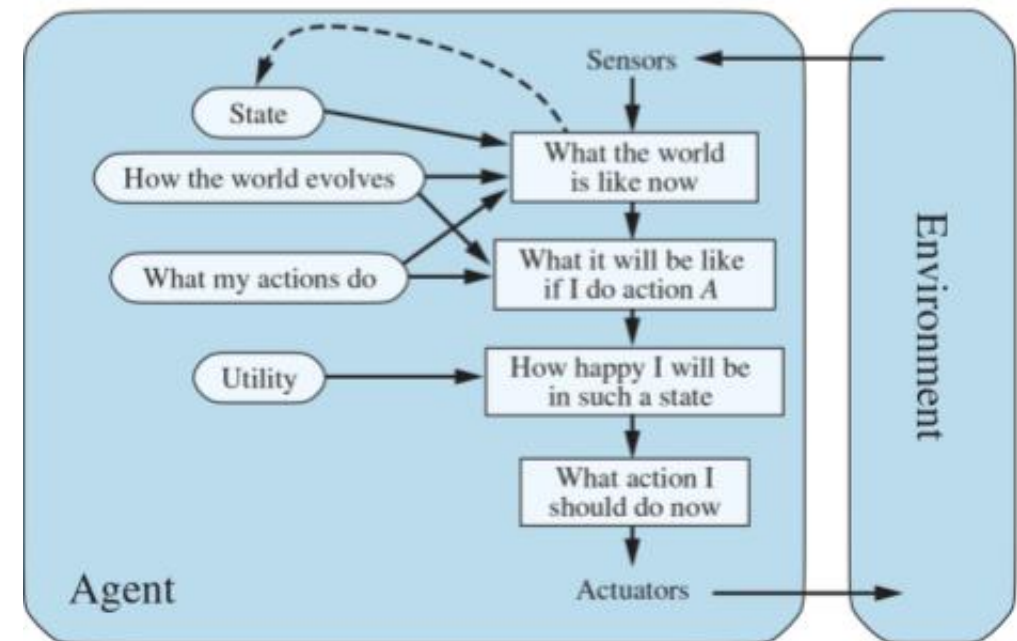
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Utility-Based Agents

- Partial Observability and nondeterminism are part of real world.
- Decision making mostly under uncertainty.
- Rational Utility-Based Agent chooses the action that maximizes the expected utility of the action outcomes.
- The utility the agent expects to derive, given the probabilities and utilities of each outcome.



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Utility-Based Agents

- Seems trivial to build Utility-based agents?
 - A utility-based agent has to model and keep track of environment.
 - Perception, Representation, Reasoning and Learning
 - Maximizing utility is also extremely difficult task.



Environments and Agents



- Environment Categories
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 - Simple Reflex Agents
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 - Utility-Based Agents
 - **Learning Agents**

Learning Agents

- Alan Turing came up with the idea that we should build learning machines and then teach them.
- Now learning is mostly a preferred method for creating State-of-the-art AI systems.
- Learning agent can be divided into four conceptual components:
 1. Learning element
 2. Performance element
 3. Critic
 4. Problem Generator



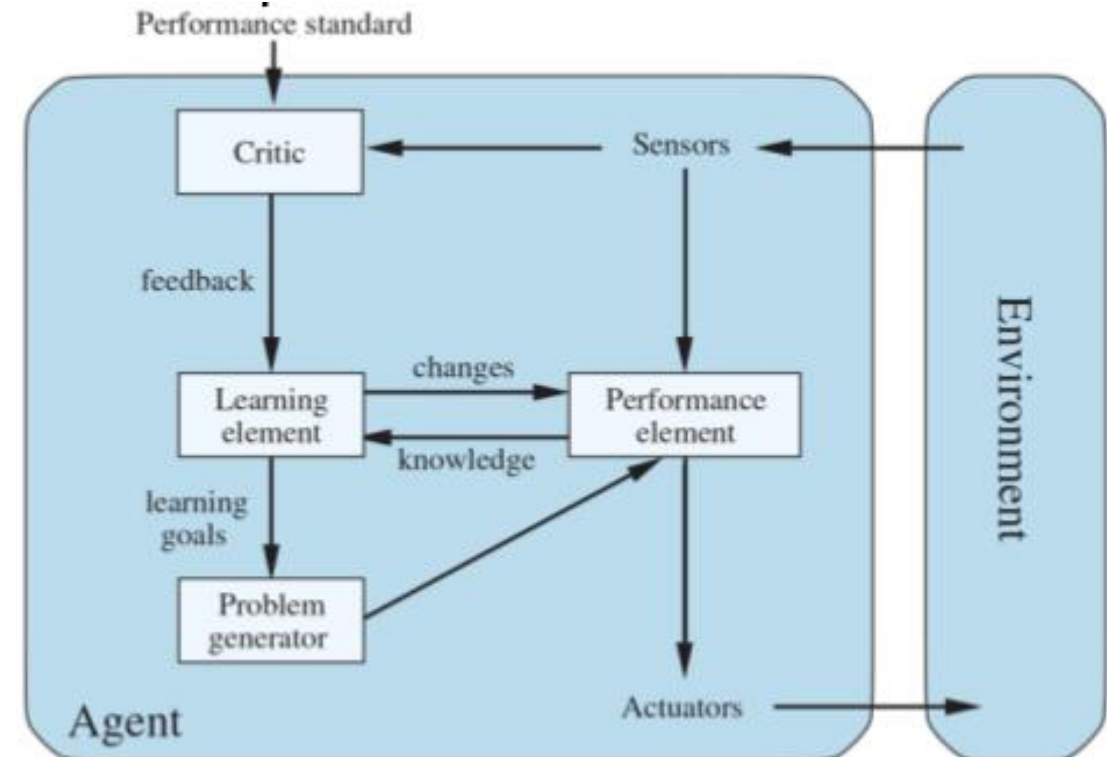
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Learning Agents

- Learning element is responsible for making improvements.
- Performance element is responsible for selecting external actions. Similar to previous agents which take in percepts and decides on actions
- Learning element uses feedback from critic on how the agent is doing and determines how the performance element should be modified to do better in future.
- Problem Generator: It is responsible for actions that will lead to new and informative experiences.



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Learning Agents

- If the automated taxi exerts a certain braking pressure when driving on a wet road, then it will soon find out how much deceleration is actually achieved and whether it skids off.
- Problem generator identifies certain parts of the model that are in need of improvement and suggest experiments.
- Improving the model components so that they conform better with reality.

